Urban Green Infrastructure and Ecosystem-Service Deficit in Tartu, Estonia

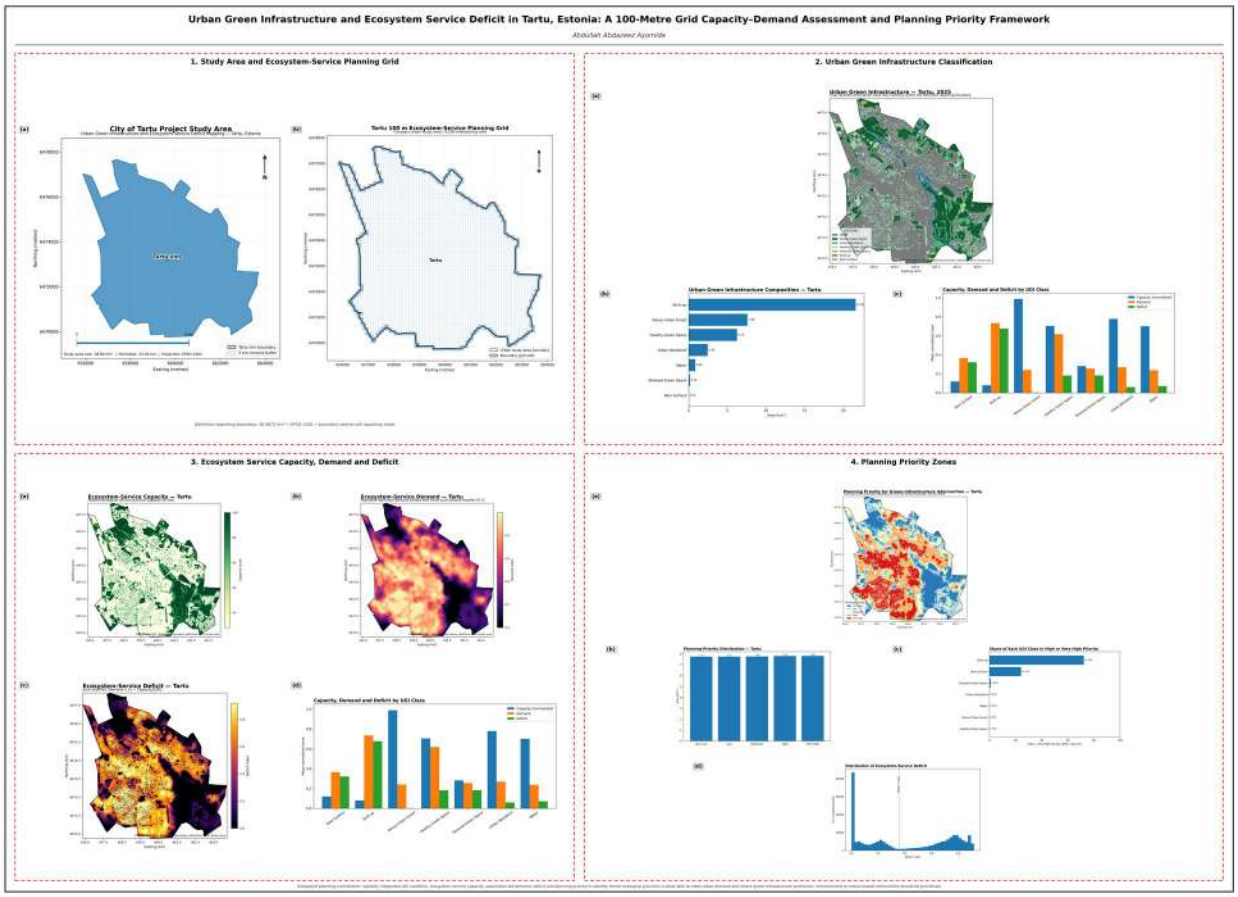
A spatial capacity-demand assessment for green-infrastructure planning

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Executive Summary

This project examines where local urban green-infrastructure capacity is weak relative to ecosystem-service demand across Tartu. Sentinel-2, Dynamic World and WorldPop were integrated on a common 10 m grid to classify urban green infrastructure, estimate relative capacity, model demand, calculate deficit and translate the result into planning-priority classes.

The reporting area is 38.8872 km². Built-up land covers 21.6040 km² (55.56%), while the four vegetated UGI classes together cover 16.4325 km² (42.26%). Mean capacity is 41.54/100, mean demand is 0.5768 and mean deficit is 0.4110. High and Very High planning-priority areas cover 15.6229 km² (40.17%) of the reporting area, and 72.30% of Built-up land lies in those two priority classes.



Project overview: green infrastructure, ecosystem-service deficit and planning priority in Tartu.

1. Planning Question

Where does demand for urban ecosystem services exceed the local capacity of green infrastructure in Tartu, and which areas deserve the most planning attention?

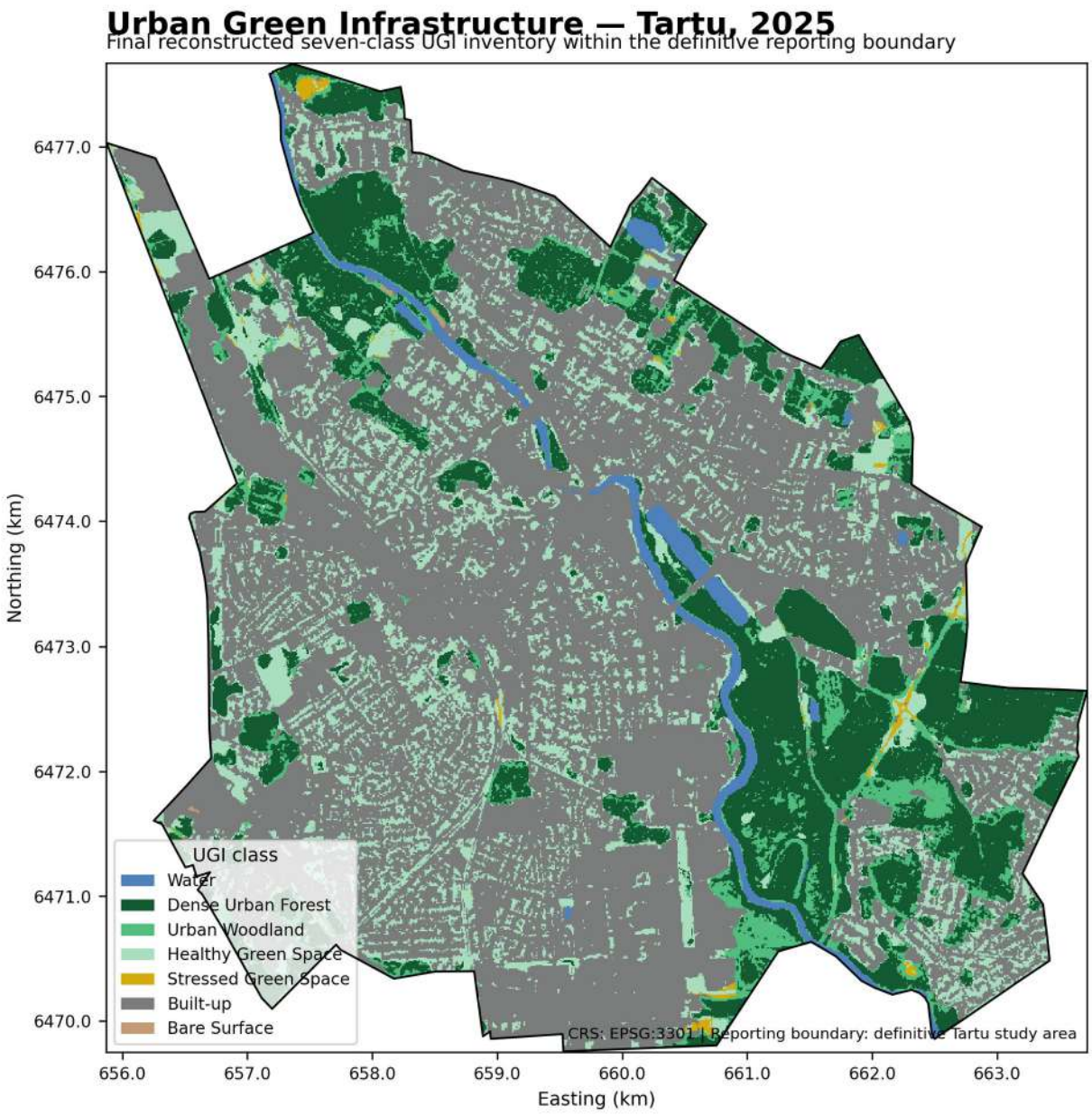
2. Study Area and Data

The final reporting area covers 38.8872 km^2 in Tartu, Estonia. The analysis uses EPSG:3301 and a 10 m reporting grid. Sentinel-2 provides spectral condition indicators; Dynamic World provides land-cover identity; WorldPop 2020 supplies the dominant population-demand proxy.

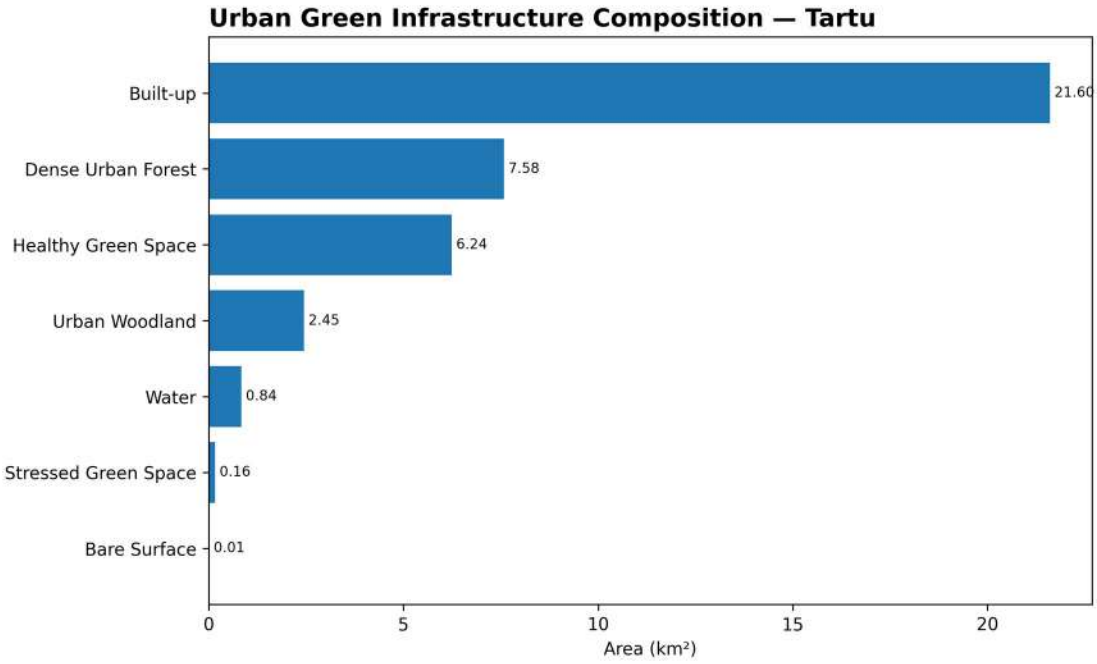
The Sentinel-2 index stack contains NDVI, EVI, NDMI, NDBI, MNDWI, SAVI and BSI. These indices are used together with land-cover information to distinguish seven urban land and green-infrastructure classes.

3. Urban Green Infrastructure

The final classification contains Water, Dense Urban Forest, Urban Woodland, Healthy Green Space, Stressed Green Space, Built-up and Bare Surface.



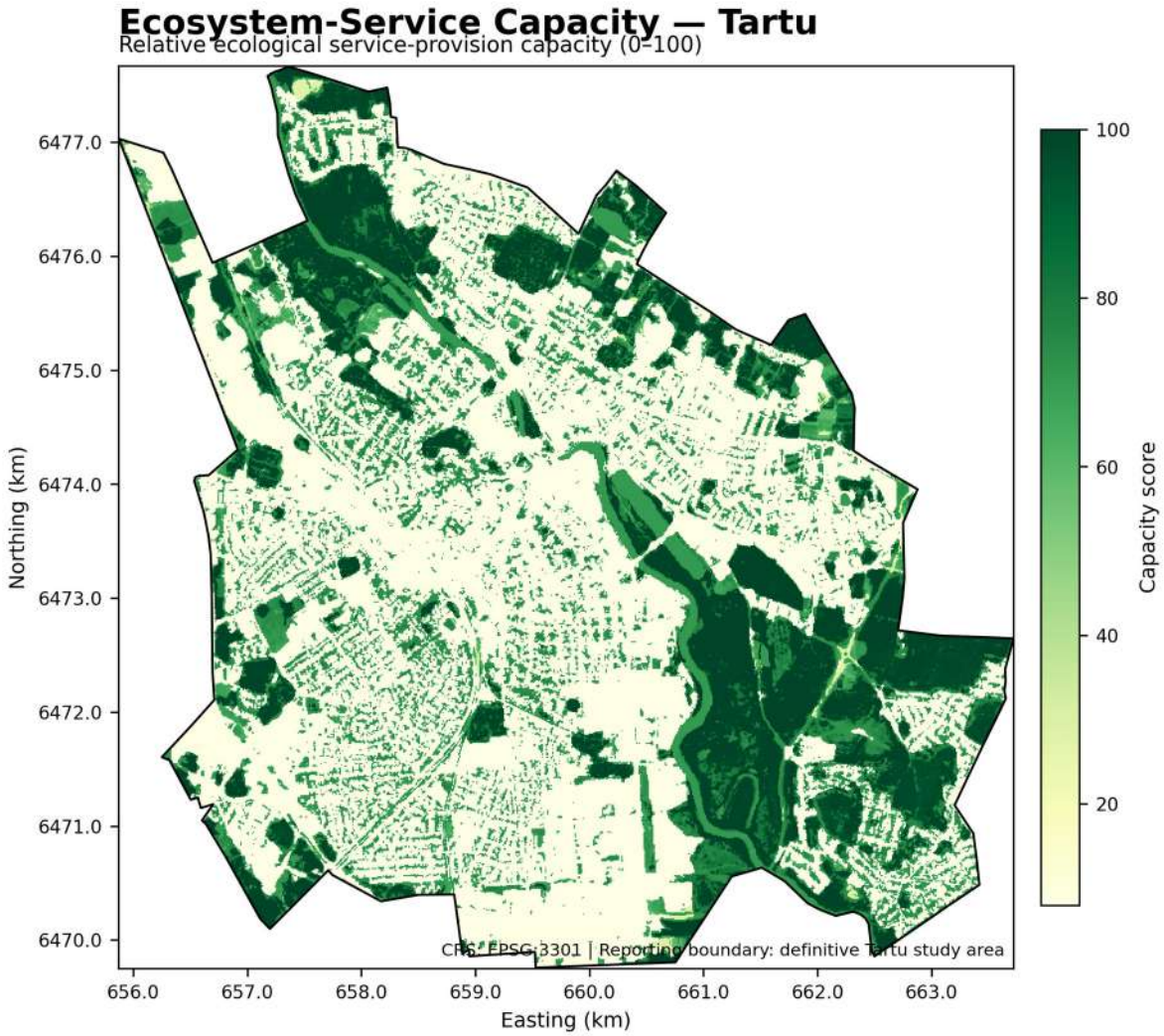
Urban green-infrastructure classification.



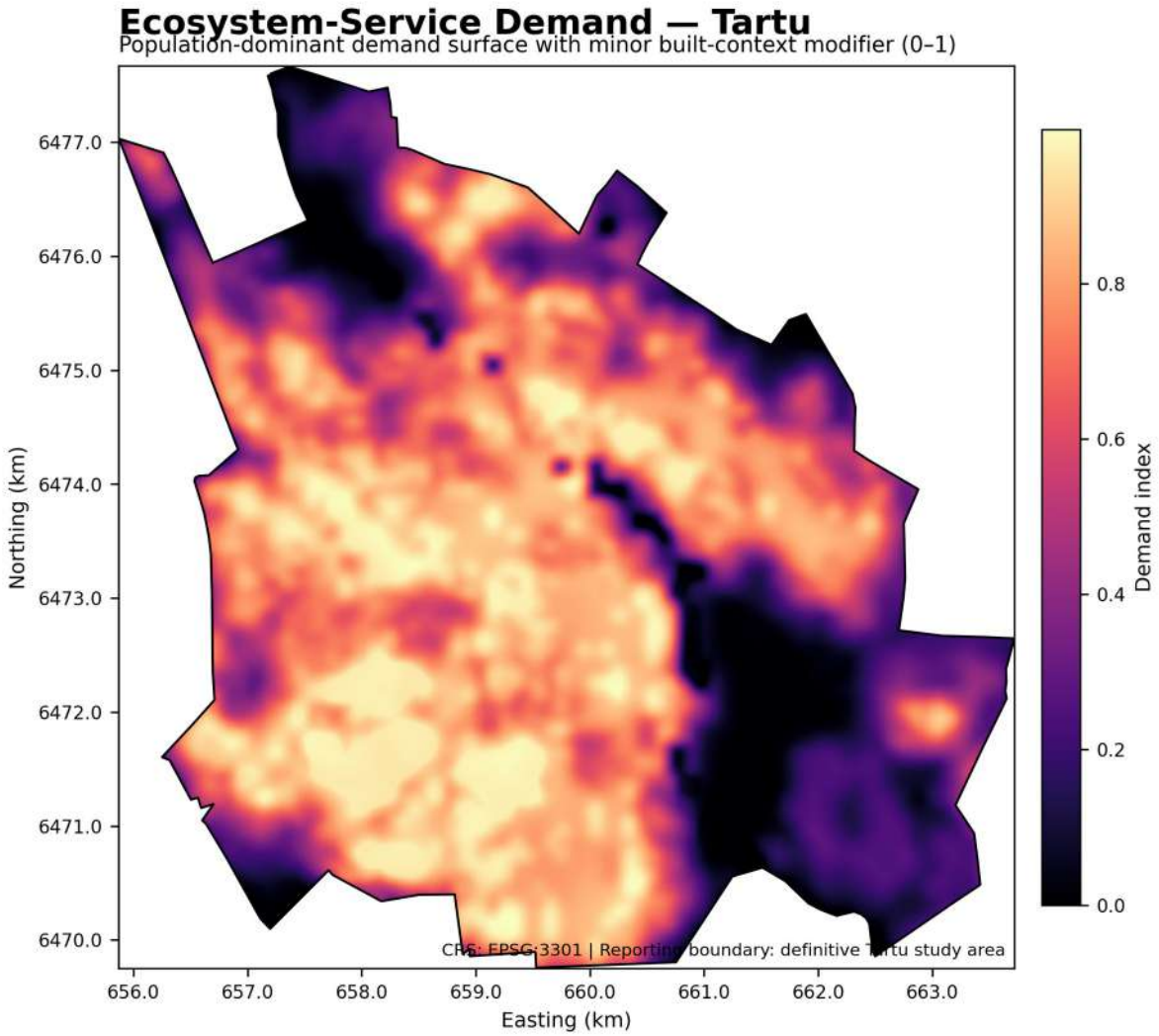
Area distribution by UGI class.

4. Ecosystem-Service Capacity and Demand

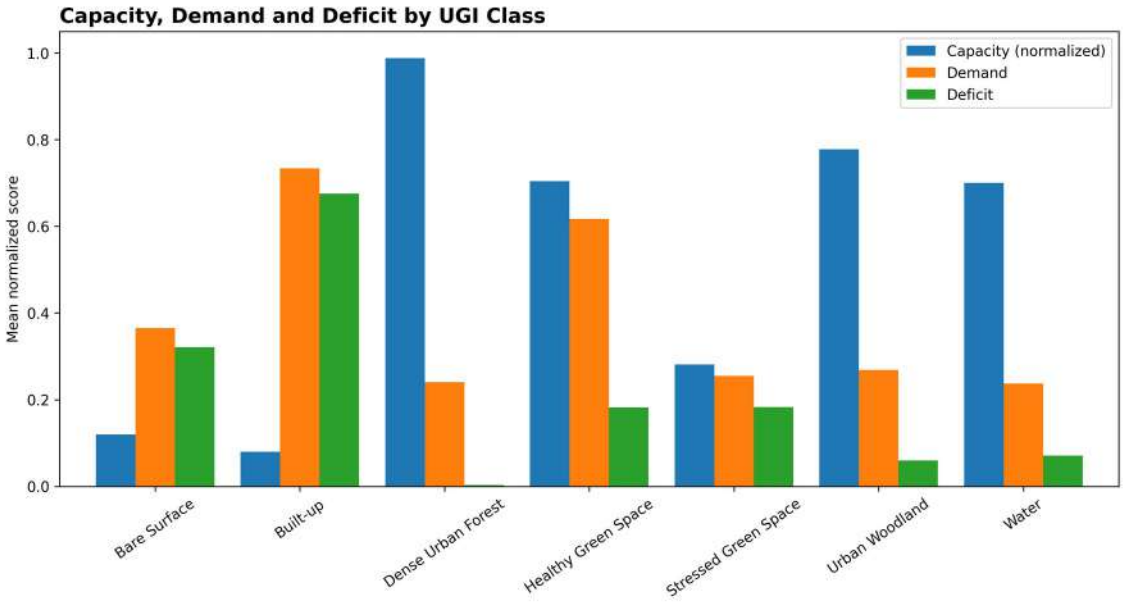
Capacity is expressed on a relative 0-100 scale. Vegetated classes receive higher baseline capacity, refined by vegetation condition. Demand is modelled mainly from WorldPop, with built context used as a smaller modifier. Capacity and demand are constructed independently before the deficit is calculated.



Relative ecosystem-service capacity.



Relative ecosystem-service demand.

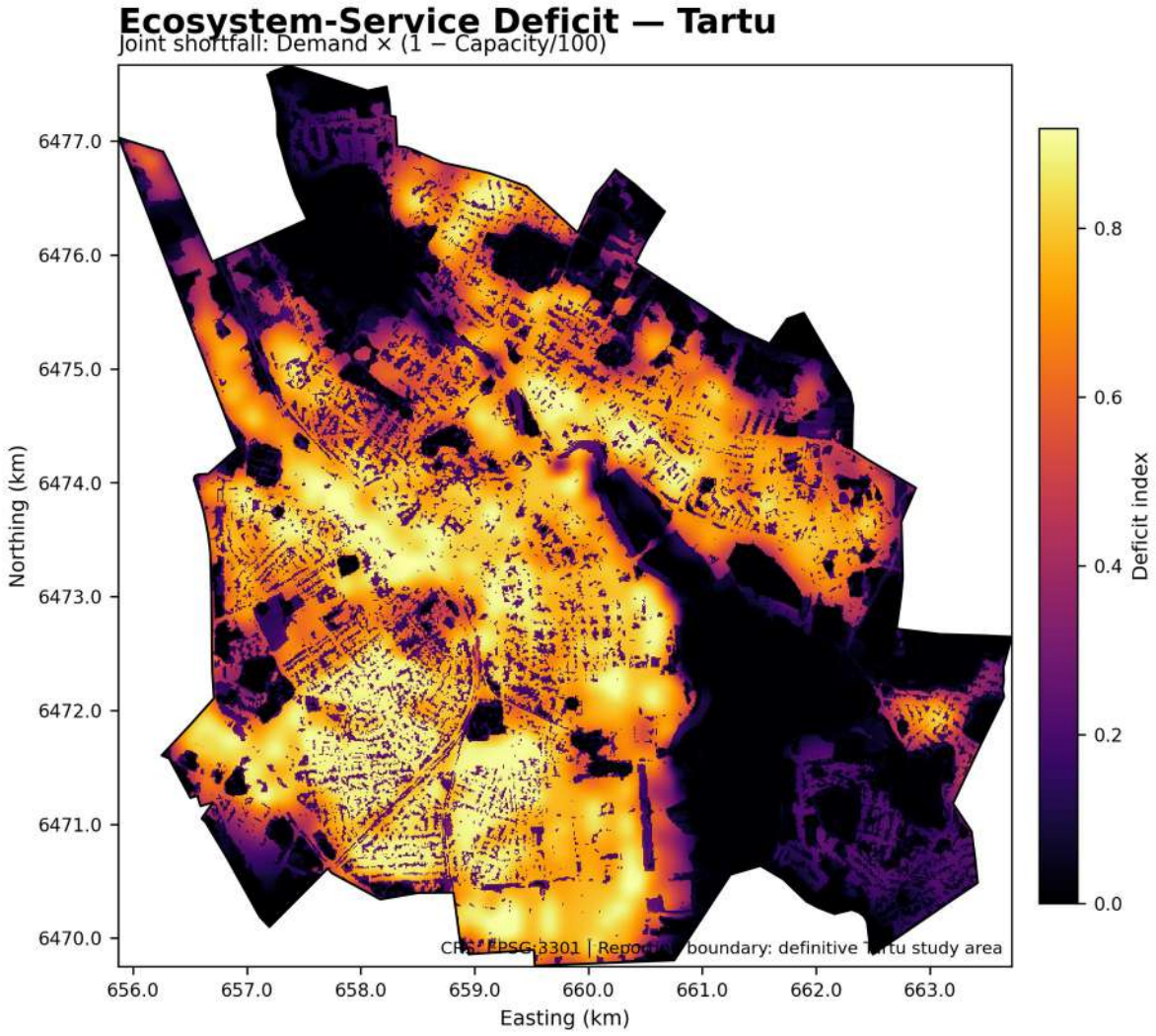


Mean model components by UGI class.

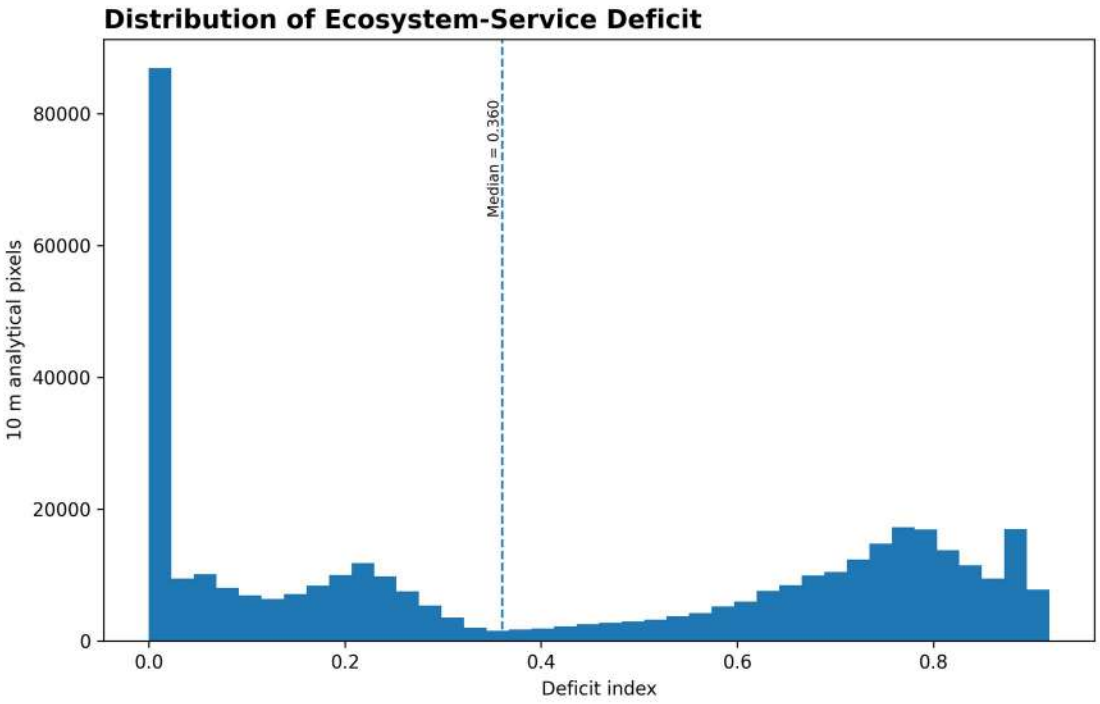
5. Ecosystem-Service Deficit

Deficit is calculated as Demand × (1 - Capacity/100). This produces high deficit where demand is comparatively high and local ecological capacity is low.

The overall mean deficit is 0.4110. Built-up land has mean capacity 8.00/100, mean demand 0.7339 and mean deficit 0.6751. Across the reporting area, demand and deficit are positively associated (Spearman ρ = 0.8715), while capacity and deficit are negatively associated (ρ = -0.8764).



Ecosystem-service deficit surface.

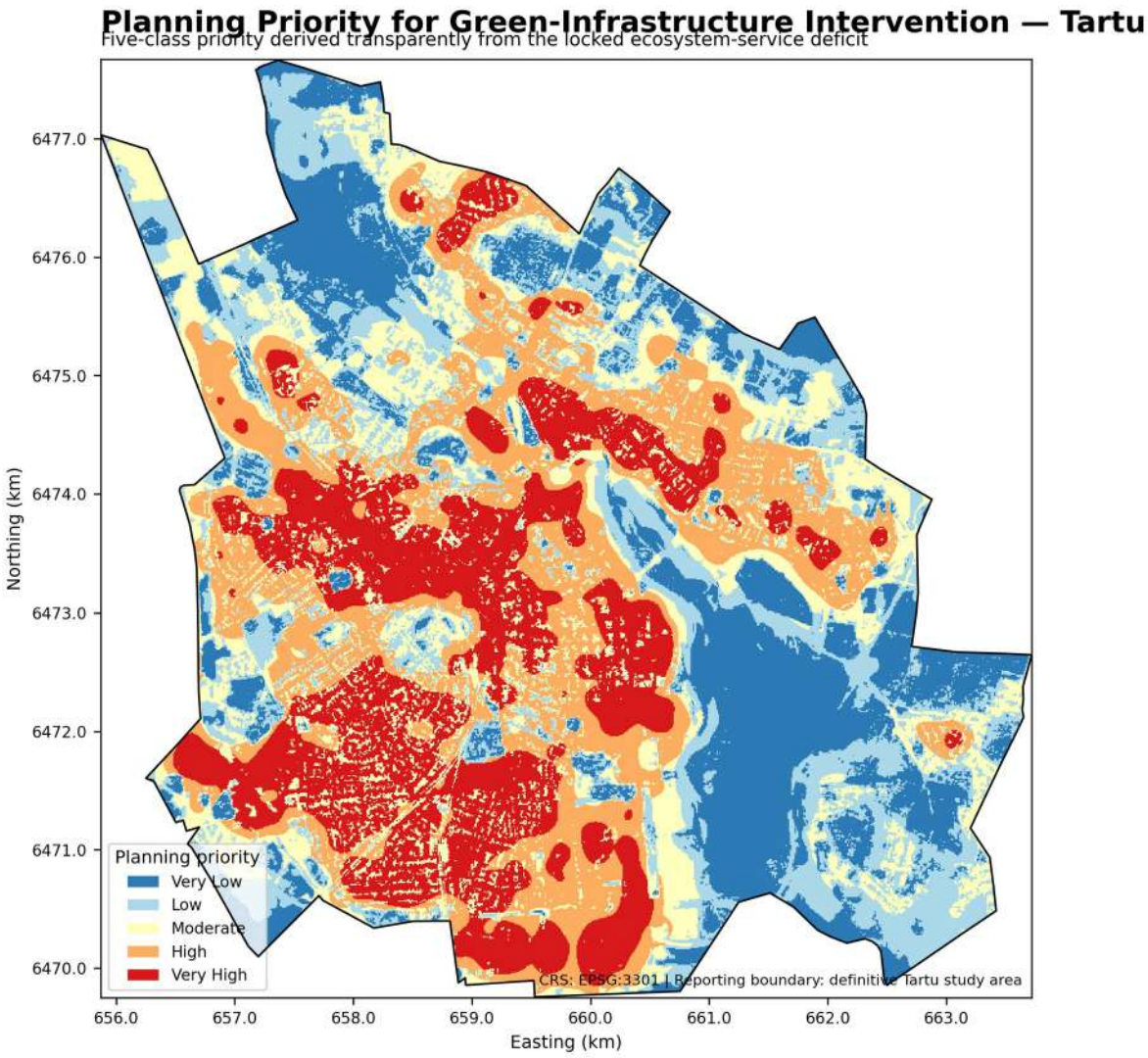


Distribution of ecosystem-service deficit.

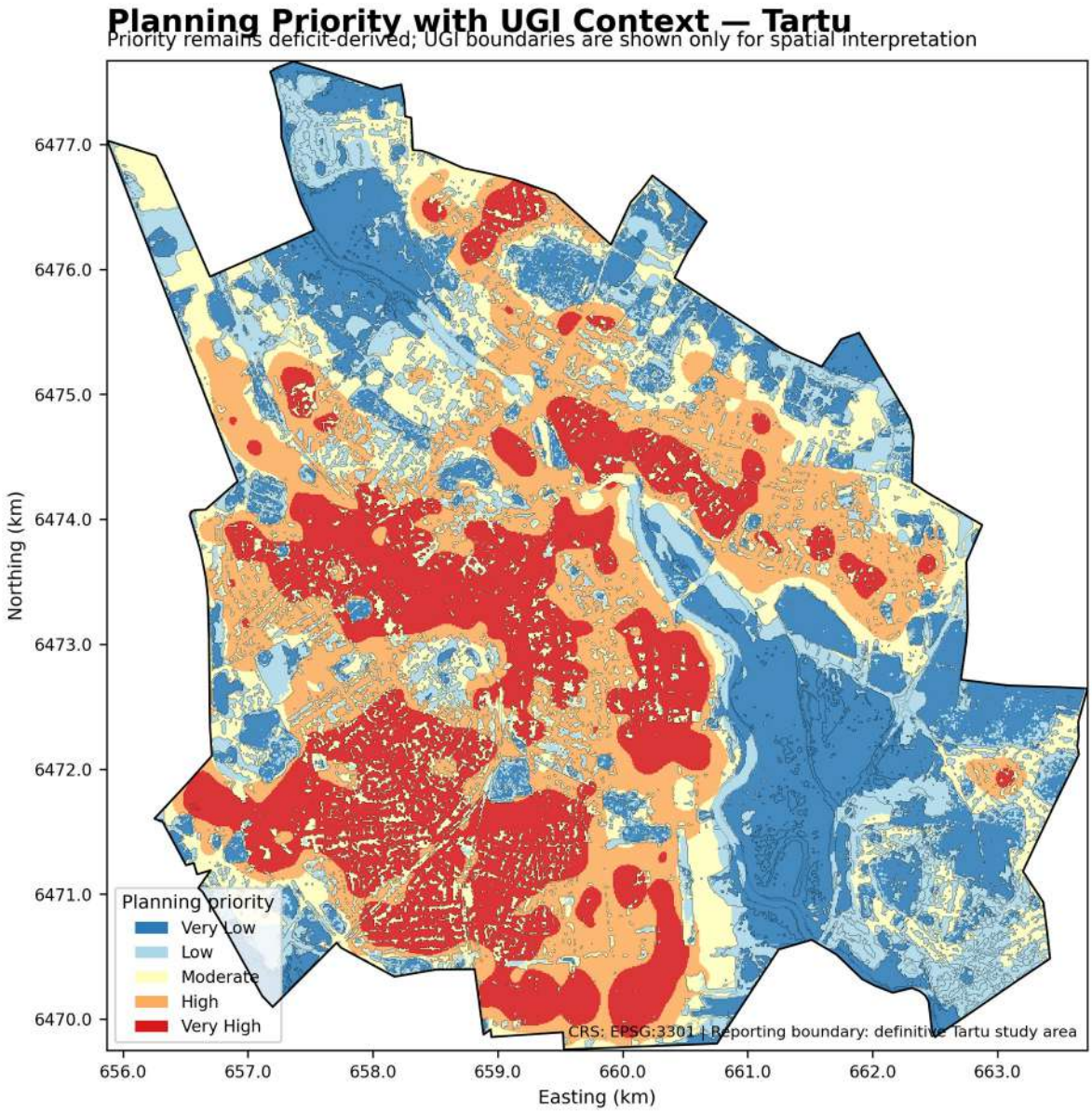
6. Planning Priority

The continuous deficit surface is translated into five relative planning-priority classes: Very Low, Low, Moderate, High and Very High. High and Very High classes together cover 15.6229 km² (40.17%) of the reporting area.

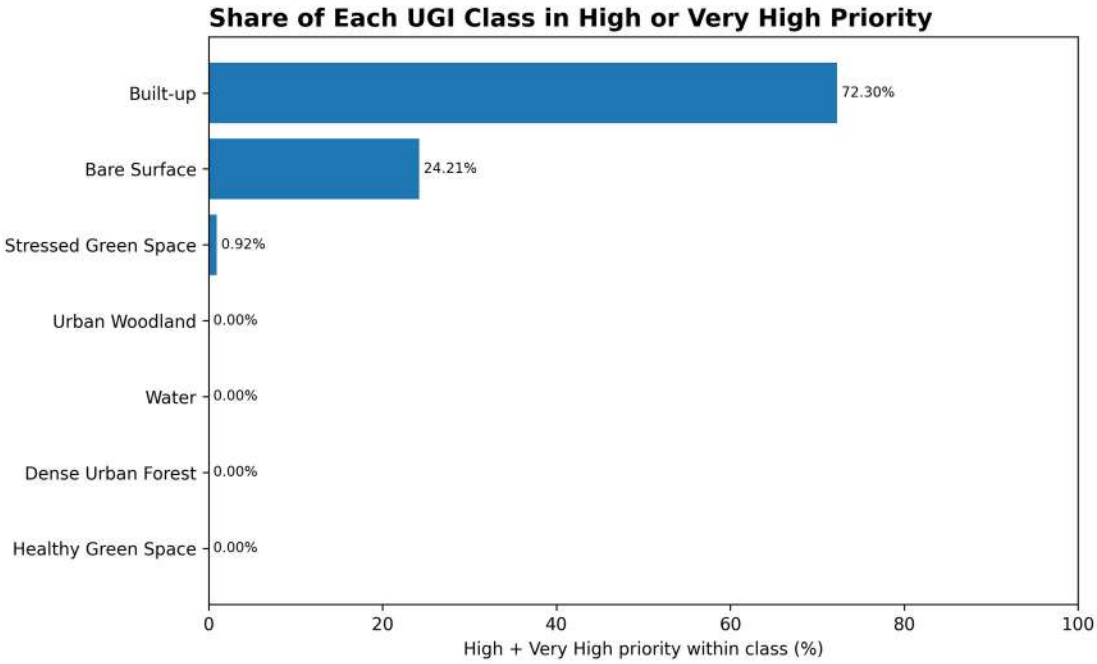
The clearest planning concentration is within the built-up fabric: 72.30% of Built-up land falls in High or Very High priority. Deficit and planning priority are strongly associated (Spearman $\rho = 0.9802$), which is expected because priority is derived from the deficit ranking.



Five-class planning-priority surface.



Planning priority shown with UGI context.



Share of each UGI class in High or Very High priority.

7. Planning Value

The results show that Tartu's challenge is not simply the quantity of green land but the spatial mismatch between ecological capacity and concentrated urban demand. High-priority urban areas are suitable for closer assessment of interventions such as street-tree expansion, pocket parks, courtyard greening, green roofs and walls, permeable surfaces, stormwater-oriented green infrastructure and improved pedestrian connections to existing green assets.

Existing high-capacity forest and woodland should also be treated as assets to protect from unnecessary fragmentation.

8. Limitations

Capacity, demand and deficit are relative planning indices. They are not direct measurements of cooling, runoff retention, carbon sequestration, recreation or health outcomes. WorldPop 2020 is a demand proxy rather than household-level population data, and alignment to a 10 m grid does not make it a 10 m census.

Priority classes are relative to Tartu and should not be transferred directly to another city without recalibration. Site-level decisions still require information on land ownership, public access, utilities, costs, design constraints, local needs and field conditions.

9. Tools and Data Sources

Tools: Python, Rasterio, GeoPandas, Pandas, Google Earth Engine, QGIS/ArcGIS and remote-sensing workflows. Data: Sentinel-2 Surface Reflectance, Dynamic World, WorldPop 2020 and the Tartu study boundary.

10. Citation

Abdullah Abdazeez Ayomide (2026). Urban Green Infrastructure and Ecosystem-Service Deficit in Tartu, Estonia.